

Making a 20-Year-Old House Work

Moving into an older home is often accompanied by unhappy surprises affecting energy use and comfort. The solution requires dealing creatively with a number of details.



by **Larry Kinney**

Almost five years ago, my wife, Wyncia, and I decided to move from Syracuse, New York, to Boulder, Colorado. Wyncia spent a December week with a very patient real estate agent searching for a home. She looked at dozens, e-mailing back pictures and descriptions to me each evening. We finally decided on the first one she saw: a 20-year-old, 2,060 ft² passive-solar dwelling that rests on the side of a mountain at 7,800 feet above sea level. It came with an active solar water heater, plenty of space for the dog to run around, an attached 610 ft² garage, and a beautiful view of the valley below.

"It's the only house I saw that felt comfortable on windy days," Wyncia reported, "and the garage is big enough for a shop." This good news was accompanied by some bad.

Electric-resistance radiant heaters heated the house; a local thermostat controlled each one. Boulder is in coal country and electricity is comparatively cheap (7.5 ¢/kWh), but I've always regarded heating with electricity as inelegant. Better to use it for running power tools, lights, and stereo systems. I had my friend Rob deKieffer, a Boulder native and longtime energy buff, inspect the home and tell me how it worked. He found a frozen pipe, a window that didn't crank, and several outlets whose voltage dropped by a bit over 5% when exposed to a 15-amp load, but he pronounced the place in good shape. So we signed papers the day before the owner received an offer of \$10,000 more than we paid. It was good timing—and a good omen.

The home's first owner, who was also the designer, did a whole lot that was almost right—but not quite. For instance, it's fairly obvious that he had a hard time clearly defining the home's envelope. The walls are 2 x 4 on 16-inch centers with fiberglass insulation and 1 inch of high-R sheathing. The vaulted ceilings and the attic ceiling are insulated with 10-inch fiberglass batts. So is the attic floor. And there's a hinged door made of plywood with a fiberglass batt stapled to it on the floor of the attic that swings up for the winter to block the attic vent.

There's an 8-ft-deep solar space that is a slab-on-grade with dark tile on top—a great idea for our climate, but this one definitely needed tweaking. All of the house, save for this solar space, is built over a pair of 4-ft-high crawlspaces covered with 6-mil poly. There's a two-speed,

whole-house fan with 30-inch-diameter blades mounted in the floor of the attic.

Here are a handful of things I've done to make the house work a lot better.

Making It Tighter

The initial blower door test showed 2,250 CFM at 50 Pa. Not too bad, but I wanted it tighter. The principal leaks

packing material I stapled to the surface of the shutter. I also put in a new attic hatch that fits snugly in the opening and rests on fat foam weatherstripping that has a sticky back, but is also stapled in place. (Belt and suspenders is best with these products, whose stickiness diminishes with time in dry climates like Boulder's.) I also put 2 inches of rigid insulation on the attic hatch.



The light aluminum louvers in the foreground open when the whole house fan immediately above them comes on. The insulating shutter (right) hinges down in the summer, up in the winter where it is snugly fastened with a wing nut. The new attic hatch is in the background.

were at the edges of the wood covering the sloped ceiling in the main room and solar space, at the attic fan opening, at the attic hatch, in a funky air-moving system in the solar space that didn't work, and around the back door in the downstairs bath. A few tubes of clear caulk took care of the ceiling spaces. The attic fan problem was more interesting. The fan uses a set of thin aluminum louvers to close the opening when the fan is off, but it leaked like a sieve, particularly during windy conditions. So I built an insulating shutter consisting of a wood frame enclosing 2 inches of Styrofoam blue board, and used a piano hinge to attach it to the wall of the laundry room next to the opening. It's behind the laundry room door and out of the way during the summer, but in winter it hinges up and attaches with a wing nut that tightens up the silvered

The back entrance had a poorly fitted interior door that suffered from both conductive and convective heat losses. I replaced it with an insulated door that had a broken window in it; I got it for 15 bucks at the local recycle yard. I took out the old window and replaced it with a double-hung, insulated vinyl window with a screen that began life as a salesman's demo unit. It was a bit short for the opening, so I filled in the opening with rigid insulation and covered it with some decorative tiles from Mexico, held in place with recycled redwood. I also pulled the siding off above the door and replaced rodent homes and bees' nests with urethane foam. So now we have an air sealed downstairs bathroom with daylighting, ventilation, and a functional door for a total cost of \$50 plus a weekend of labor.

Solar Space

The solar space runs for 26 ft along the back of the house and is 8 feet wide on the first floor with a sloping roof that meets the main house at the top of the



The renovated back door is insulated, sealed, and supplies ventilation when needed and natural light—unlike the interior door it replaced.

second floor. The roof provides a 1 1/2-foot overhang over the 100 ft² of south-facing clear insulated glass on the first floor. The glass section consists of six fixed panels measuring 13.2 ft² each and six 3.5 ft² windows that swing out immediately below the fixed panels. The wood has held up well and the windows are tight and well sealed.

That's the good news. The bad news is that the original owner took it upon himself to install a funky fan-and-duct system. The system pulled air from halfway up the solar space, transported it via two 30-ft-long ducts to the crawlspace under the north side of the house, and dumped it there! No opening to the living space, no nothing. So I took

out the fan, sealed and insulated an opening to an outside louver used to add fresh air, and sealed the entrances to the ducts. This left a substantially simplified passive-solar space that communicated to the rest of the house via two openings on the first floor: a sliding glass door from the dining room and a crank-out casement window above the kitchen sink. These provided about 18 ft² of opening to the solar space, but since the top of the space was sealed, and warm air rises, air movement to the rest of the house was modest.

The retrofit I undertook was designed not only to solve the solar thermal distribution problem, but also to provide some daylighting and an inside clothes-drying line. The second-floor bathroom was adjacent to the inside wall of the solar space, so I opened the wall above the tub and installed a piece of sliding Plexiglas. This provided light from the solar space and a 2 ft² opening when desired. I also opened the wall between the master bedroom and the solar space on the second floor, installed a door with glass panels, and put in an 8 ft x 8 ft platform. Thus, between the openings in the bathroom and the bedroom, there's about 20 ft² through which air from below can thermosiphon from openings of about the same size on the first floor. I also put in a pair of clotheslines on pulleys at the top of the solar space that are accessible from the new platform outside the master bedroom. This is handy, because the laundry is on the second floor. Finally, I also installed an operable skylight in the ceiling immediately above the platform. This allows for viewing the moon from the comfort of our bed, provides a bit more passive solar, and can supply additional ventilation when needed.

Boulder is blessed with about 300 days of sunshine a year, a resource most welcome to those of us accustomed to enduring cloudy winters in upstate New York. On winter evenings, we close off the solar space until the next morning, opening it up at the top and bottom as soon as the temperature of the solar space exceeds that of the rest of the house. Most days, this is around 9 a.m., although after particularly cold nights it's closer to 10 a.m. On clear winter days, the space routinely stays warmer than the rest of the house, getting to the high



The new balcony in the solar space allows access to the clothes line and to the skylight, which can be opened to provide ventilation.

80s even with the two interior doors and windows open. More important, it thoroughly heats the rest of the house. The laundry at the top of the solar space is dried by the movement of warm air, which in turn is humidified by the wet clothes. This is quite welcome in Boulder's dry climate, but may be less so in climates with lots of precipitation.

Supplemental heat for the whole house is supplied by an airtight Jotul wood stove. We haven't turned on an electric-resistance heater for three years. Instead, we burn oak purchased from a neighbor down the hill at the rate of half a pickup truckload per year—about \$40 worth. We cool with the whole-house fan on low speed and control the fan with a wind-up timer so that it turns itself off in the middle of the night.

Total space conditioning and hot water cost for the year amounts to around \$50. Water heating is completely active solar; we've never had to use the backup electric heater. The solar water heater uses a 1/25 hp pump motor that I estimate runs about 1,200 hours per year, consuming 36 kWh and costing \$2.70. The fan motor consumes 200 watts on low power, and we run it for at most 500 hours per year. Over the year that fan consumes 100 kWh, costing \$7.50.

Reviving the Garage

House space—not to say indoor air quality—is way too precious to waste in accommodating rolling stock, so we park cars outside where they belong. Two summers ago, we converted the garage on the north side of the house into a 250 ft² pottery studio and a 360 ft² shop. The first step was to cover the existing cement floor with 2-inch-thick tongue-and-groove Styrofoam blue board covered with 1/2 inch of oriented strand board (OSB), also with tongue and groove. We used low-VOC building cement between the floor and the blue board, and between the blue board and the OSB, and then stapled the OSB to the insulation. This floor system works well and is easy on the feet.

We then took down the 16-ft-long overhead door and recouped it for shelves and siding. The door opening was replaced with a wall (2 x 6 on 24-inch centers) and an insulated door I trash-picked and rebuilt. (Boulder is a nice place for such things.) We put in a pair of small, fixed insulating skylights; two recycled 7 1/2 ft² insulating windows on the north wall; and a 2.6 ft² window on the west. (It's the only west-facing



(left) The pottery is nicely daylit, so most days of the year no lights are needed. (right) The shop is designed for metal and wood work, but also includes an electronic bench. There's lots of storage above.



This space that used to be a large garage with a 16 foot door now houses the pottery (left) and the shop (right), which are pictured above.

fenestration in the house, which, along with the overhang over the south-facing windows, mostly explains why the summer cooling load is so modest.) We then covered the walls and ceiling with 8-mil reinforced poly and blew in cellulose to high density—11 inches in the ceiling and 6 inches in the walls. I also insulated the interior wall between the pottery studio and the shop with fiberglass batts, because I wanted the flexibility of heating one space without having to heat the other. The inside walls and ceiling are finished with OSB in the shop and 1/2-inch gyp in the studio, and walls and ceilings are all painted with semigloss white enamel. This hue, in combination with skylights and strategically placed 4-

ft T-8 fluorescent fixtures with electronic ballasts, yields good lighting that keeps even old guys like me from cutting off fingers in saws.

These spaces are not heated in the winter unless I plan to work there for extended periods. However, they can be heated quickly because they are very tight and well insulated, and because—since the concrete slab is isolated from the space—they have quite low mass. Accordingly, we can open the interior doors to the inside of the house, switch on 1 kW of radiant heating, and make the shop/pottery spaces quite comfortable within an hour; it might take two hours on the coldest day of the winter.

Radon Remediation

The home had been retrofitted with a subslab depressurization system, but it only sucked from one of the crawlspaces. So I installed another 4-inch PVC pipe in the other crawlspace and hooked the two together. This solved the radon-in-the-air problem, but I suspected radon in the water, so I taped an activated carbon sensor to the bottom of the lid of the water closet of the downstairs toilet. My teenage daughter was with us then, so the toilet got flushed pretty often. Sure enough, the test came back indicating 10 picocuries per liter (pCi/l), so we had radon in the water. Accordingly, I did a more serious test with a local lab and became convinced that we should put in a treatment system. Wyncia and I did the plumbing and used the old fan-and-duct cabinetry in the solar space to house some fancy water treatment hardware installed by a professional. It seems to work and the water tastes good!

Finishing Touches

The solar water heater has been on the house for 20 years and works quite well. The only change I've made is to lower the two thermostats on the backup heater to 110°F; as far as I can tell, they've never switched on the electric elements. We also installed low-flow showerheads, an aerator at the kitchen sink, and a more efficient well pump.

Snug Home

The Snug Home is a construction technique aimed at producing very energy-efficient structures whose labor and materials costs are substantially below those of most homes—roughly \$50/ft². Instead of a poured-concrete foundation, these structures are supported by pressure-treated wood poles placed on 8-ft centers in 5-ft-deep holes with a concrete plug in the bottom—in much the same way as pole barns are supported. Trenches 4 ft deep are dug between the poles to accommodate 4-inch to 6-inch-thick closed-cell insulation, such as Styrofoam blue board. The perimeter insulation is used to define the edges of a concrete slab and mate with wall insulation without thermal breaks. Two-by-four framing members are installed on their edges horizontally on 2-ft centers inside and outside the poles to form a 12 1/2-inch-thick wall. A fiber-reinforced 8-



This is the first Snug Home that was built in Morrisville, NY. It was designed by Paul Howells, an engineer with the Syracuse Research Corporation.



This shed is built like a Snug Home and never goes below freezing, even without heating.

mil plastic vapor barrier is installed next to the inside surfaces of the poles. It is mated with the below-slab and ceiling vapor barriers via overlapping plus stapling, and silicon-based caulk is used to achieve a continuous seal. OSB is glued and nailed to the outside of the structure. The siding of choice (for example, shingles or stucco) is then installed. Cellulose insulation is blown to high density into the space between the vapor barrier and the OSB, as well as into the attic. Plumbing, wiring, and ducts for an air-to-air heat exchanger are installed in the 3 1/2 inch wall spaces on the inside of the vapor barrier. Then this space is insulated with R-11 batt insulation, and drywall or other interior finishing material is installed. High-quality windows are used to ensure solar gain in the winter, and both overhangs and exterior shading are used to limit summertime heating.



Raised beds with compost-rich soil help growth and the framework with netting impedes deer from having salad before human beings have a chance at it.

The refrigerator that came with the house used 1,400 kWh per year, so we replaced it with a 21 ft³ unit that uses 525 kWh per year. (Both figures are from

long-term measurements.) We also bought a chest-style freezer to accommodate vegetables from a local community-supported agriculture organic farm. The

freezer lives next to a wall of the shop and uses 220 kWh per year.

Instead of using the giant electric oven in the electric stove, we bought a combination microwave/convective broiler that serves us quite well, save for Thanksgiving Day. We also got some pots and pans whose bottoms are very flat, so the heat transfer between the glass stove top and the pots and pans is as high as is practical. This ensures good conductive—instead of just radiative and convective—heat transfer. We installed more than 20 CFLs and put dimmers on the few incandescents we still have. The house was full of light cans, most of which were black on the inside. Since that seemed needlessly inefficient to me in terms of both heat and light, we took the cans apart, sprayed the insides with semigloss white enamel, and then reinstalled them.

The result is a very tight, well-insulated structure—R-22 perimeter insulation, R-42 wall insulation, and R-50 ceiling insulation—that is coupled to the earth. Deep earth temperatures approximate the annual mean of air temperatures; in the southwestern United States, these range from 48°F to 67°F. Earth-coupled structures such as the Snug Home resist going below deep earth temperature even during extended cold spells in the winter, and their summertime performance is substantially enhanced as well. This is because the earth itself is somewhat warm and functions as thermal mass. Perimeter insulation ensures that the portion of the earth close to the surface, whose temperature varies with the season, is effectively outside of the thermal envelope. A temperature bubble gradually builds up in the earth under the slab. As a result, wintertime energy performance in the home's second year is even better than its first-year performance.

To experiment with the technique in a dry, mountainous climate, colleagues and I



When it rains, the shed's roof dumps water into a barrel that is used to water plants during dry periods. A vine (Virginia creeper) that will hopefully cover the south-facing glass during the summer by next year is off to a good start.

built a test structure (that doubles as a storage shed) on our property, where deep earth temperature is approximately 48°F and base 65 heating degree-days are about 6,500. Built according to Snug Home principles with only modest fenestration (9 ft² of windows, facing south), the structure

was not heated at all during the last two winters (except for a brief period in December 2002, when we heated it to collect performance data). During the coldest period of the year, when outside temperatures were at -5°F, the inside temperature never descended below 36°F. In mid-December 2002, the structure was heated to 68°F by a 900W thermostatically controlled space heater while we tracked inside-outside air temperature. After reaching steady-state conditions, the structure consumed only 1 Btu/ft²/HDD. This is about 15% of the consumption of conventional new homes in the southwestern United States and is consistent with results achieved by Snug Homes in the more severe climate areas of New York State.

The Snug Home uses cellulose insulation—recycled newsprint that has been treated to neither burn nor be attractive to vermin, while providing good karma for humans. A key advantage of the Snug Home technique is that the structure can be built by amateurs at a fraction of the labor required to build a straw bale home.

What's Left?

Plenty is left, of course. My next major project is to design some insulating shutters for the outside of the main windows in the solar space. At present, the solar space loses 76 Btu per hour/°F, 65% of which (50 Btu per hour/°F) is lost through the windows alone. Covering the window wall with exterior insulating shutters at night will lower losses through the window system to 7 Btu per hour/°F, bringing overall losses from the solar space to 33 Btu per hour/°F. Instead of closing off the space and allowing it to cool overnight, we could easily use the wood stove to make up for this loss, which would amount to only 1,650 Btu per hour on a night whose average temperature is 20°F. Not closing off the space would enable it to start contributing to heating the house within

an hour of sunrise. I'm also interested in automating the opening and closing of the shutters with a simple system that senses both insolation and temperatures to optimize energy performance.

The other project on the boards is to use solar energy to heat the hot tub. Although it's well insulated with urethane and has a thick insulating cover, I'm ashamed to admit that the hot tub is the greatest remaining source of sin—at least in regard to energy use! I would like to install a first-surface mirror—one with no glass layer covering the reflective surface—on half of the insulating cover. (A German company manufactures a polished aluminum product that can take the hit of the weather and nonetheless reflect over 95% of the light that impinges on it.) I plan to install a pair of worm drives (with a single small dc gear motor) to raise the mirror to a vertical

position on the north side of the tub when solar conditions are right. A clear plastic cover will remain on the top of the tub to eliminate evaporative cooling and retard convective losses. Sensors will be used to optimize reflective shutter opening to save as much back-up electricity as possible, while closing the lid when the tub temperature gets too high or a high wind comes up.

When I have the electricity use sufficiently low, I'll give some thought to getting off the grid by using PV. Meanwhile, efficiency improvements are almost certainly more cost effective.



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